HW03

ENAE 641: Linear System Dynamics

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Problem 1

The file `F16.mat` on ELMS contains a standard state-space perturbation model for the longitudinal dynamics of an F16 aircraft in level flight at $100\,000\,\mathrm{ft}$ moving with speed $500\,\mathrm{ft\,s^{-1}}$. The four states of this model are: speed, angle of attack, pitch angle, and pitch rate, in that order.

a) Determine the real block modal form for the $\boldsymbol{A}$ matrix, and show that the system is stable.

b) Specify a physical (i.e. real-valued) initial condition which will uniquely excite the phugoid (slowly oscillating) mode. Repeat for the short period (fast oscillating) mode. Interpret each in terms of the physical variables.

c) The two inputs to this model are throttle and elevator deviations from trim, respectively. Which mode is most strongly excited by the throttle? Which mode is most strongly excited by the elevator?

HINT: Look at the corresponding modal $\boldsymbol{B}$ matrix for the transform in Part A

a) Answer

b) Answer

c) Answer

Problem 2

Consider a time-periodic system from HW02:

$$\boldsymbol{A}(t) = \begin{bmatrix} -1 + 1.5\cos^2(t) & 1 - 1.5\sin(t)\cos(t) \\ -1 - 1.5\sin(t)\cos(t) & 1 + 1.5\sin^2(t) \end{bmatrix}$$

This system is time-periodic, hence it admits a Floquet decomposition. Determine the components $(\boldsymbol{P}(t), \boldsymbol{Q}, \boldsymbol{R})$ of this decomposition.

Answer

Problem 3

For the linear time-varying system

$$\dot{\mathbf{x}}(t) = \boldsymbol{A}(t)\mathbf{x}(t) \quad \mathbf{x}(t_0) = \mathbf{x}_0$$

if

$$\boldsymbol{A}(t) = \begin{bmatrix} 0 & t\sin(t)\cos\left(\sqrt{t}\right) \\ -t\sin(t)\cos\left(\sqrt{t}\right) & 0 \end{bmatrix}$$

and $\mathbf{x}_0^\mathsf{T} = [-1, 2]$, find $\|\mathbf{x}(t)\|_2$ for all $t \geq t_0$. What does this tell you about the stability properties of this system (unstable, asymptotically stable, or stable)?

Answer

Problem 4

a) For the F16 model in Problem 1, find a constant, positive-definite matrix $\boldsymbol{P}$ so that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\mathbf{x}^{\mathsf{T}(t)} P \mathbf{x}(t)\right) = -\left\|\mathbf{x}(t)\right\|^2$$

(Use `MATLAB`'s `lyap()` function, but read its documentation carefully!) Verify this $\boldsymbol{P}$ is positive definite.

b) Use this $(\boldsymbol{P}, \boldsymbol{Q})$ pair to bound the exponential convergence of $\|\mathbf{x}(t)\|$ from an arbitrary $\mathbf{x}_0$.

a) Answer

b) Answer

Problem 5

a) For the F16 model in Problem 1, what are the eigenvalues of $\boldsymbol{A} + \boldsymbol{A}^{\mathsf{T}}$? Do these eigenvalues allow you to conclude the kind of exponential stability result that you obitained in Problem 1 and Problem 4?

b) Is there a contradiction between the results of Part A and that of Problem 1 and Problem 4? Explain carefully in light of Lyapunov theory.

HINT: What $\boldsymbol{P}$ are you implicitly using in the test of Part A? Should you expect an arbitrary choice such as this to satisfy Lyapunov's requirement?

a) Answer

b) Answer

Problem 6

For the homogenous dynamics of a linear system, suppose that the symmetric matrices $\boldsymbol{P}(t), \boldsymbol{Q}(t)$ satisfy the Lyapunov equation, with $\boldsymbol{P}(t)$ bounded and uniformly positive definite, and $\boldsymbol{Q}(t)$ bounded. Let $\nu(t)$ be the minimal eigenvalue of $\boldsymbol{Q}(t)$, i.e.

$$\nu(t) = \min_k \lambda_k(\boldsymbol{Q}(t))$$

Determine the "strongest" property (in order: inconclusive, stable, asymptotically stable, or exponentially stable) you can conclude for the following cases below. Show your reasoning carefully, and note that some or all of these cases might not fit neatly into the categories specifically considered in class.

a) $\nu(t) = \frac{1}{t}$

b) $\nu(t) = -ce^{-t}$ with $c > 0$

c) $\nu(t) = \frac{1}{t^2}$

d) $\nu(t) = 2 - 3\sin(t)$

a) Answer

b) Answer

c) Answer

d) Answer

Problem 7

Generalize your observations from Problem 6. What general conditions on $\nu(t)$ would allow you to conclude isL stability? Asymptotic stability? Exponential stability?

NOTE: This question is asking you to go beyond the comparatively straightforward results discussed in lecture; for example to demonstrate formally that stability can sometimes be concluded if $\nu(t) < 0$ for some or even all $t \geq t_0$, etc.

Answer

Code

./src/index.m Matlab

```
1  %[text] # ENAE 641 - Homework 3
2  %[text] Vai Srivastava
3  % clear history
4  clear;
5  clc;
6
7  % load provided data
8  addpath(genpath("../data"));
9  load("F16.mat");
10 %%
11 %[text] ## Problem 1
12 disp(A); %[output:86bf1d2b]
13 disp("foo") %[output:8a4ae795]
14
15 %[appendix]{"version":"1.0"}
16 %---
17 %[metadata:view]
18 %   data: {"layout":"onright"}
19 %---
20 %[output:86bf1d2b]
21 %   data: {"dataType":"text","outputData":{"text":"    -3.000000000000000e-02
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   -8.442895096831324e-02    -1.388160608666941e+00    -8.172788905091380e-04
   9.066014090861495e-01\n                          0                         0
   0     4.252051613688052e-01\n     1.212174017601545e-01    -4.782799599324915e+00
   0    -2.000000000000000e+00\n\n","truncated":false}}
22 %---
23 %[output:8a4ae795]
24 %   data: {"dataType":"text","outputData":{"text":"foo\n","truncated":false}}
25 %---
```